

HW_8_ColbyFrison

Homework 8

Name: Colby Frison

OUID: 113568816

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Questions

1. Problem 5-2 ~

What is the difference between a direct and an indirect address instruction? How many references to memory are needed for each type of instruction to bring an operand into a processor register?

Answer:

- **Direct Addressing:** The address field contains the effective address of the operand. Only one memory reference is needed to access the operand.
- **Indirect Addressing:** The address field points to a memory location that contains the effective address of the operand. Two memory references are needed: one to get the effective address and another to access the operand.

2. Problem 5-4 ~

For each transfer, specify: (1) the binary value for bus select inputs (S_2, S_1, S_0) and ($S\{0i\}$); (2) the register whose LD control input must be active; (3) a memory read or write operation (if needed); and (4) the operation in the adder and logic circuit (if any).

- **a. ($AR \rightarrow PC$)**
 1. Binary value: $S_2S_1S_0 = 010$ (selects PC which is connected to bus input 2)
 2. LD control: AR
 3. Memory operation: None
 4. Adder/Logic: None
- **b. ($IR \rightarrow M[AR]$)**
 1. Binary value: $S_2S_1S_0 = 111$ (selects Memory data output which is connected to bus input 7)
 2. LD control: IR
 3. Memory operation: Read
 4. Adder/Logic: None
- **c. ($M[AR] \rightarrow TR$)**
 1. Binary value: $S_2S_1S_0 = 110$ (selects TR which is connected to bus input 6)
 2. LD control: None

3. Memory operation: Write
4. Adder/Logic: None
- **d. ($AC \leftarrow DR, DR \leftarrow AC$) (done simultaneously)**
 1. Binary value: For $AC \leftarrow DR$: $S_2S_1S_0 = 011$ (selects DR which is connected to bus input 3)
For $DR \leftarrow AC$: $S_2S_1S_0 = 100$ (selects AC which is connected to bus input 4)
 2. LD control: AC, DR
 3. Memory operation: None
 4. Adder/Logic: None

3. Problem 5-7 ~

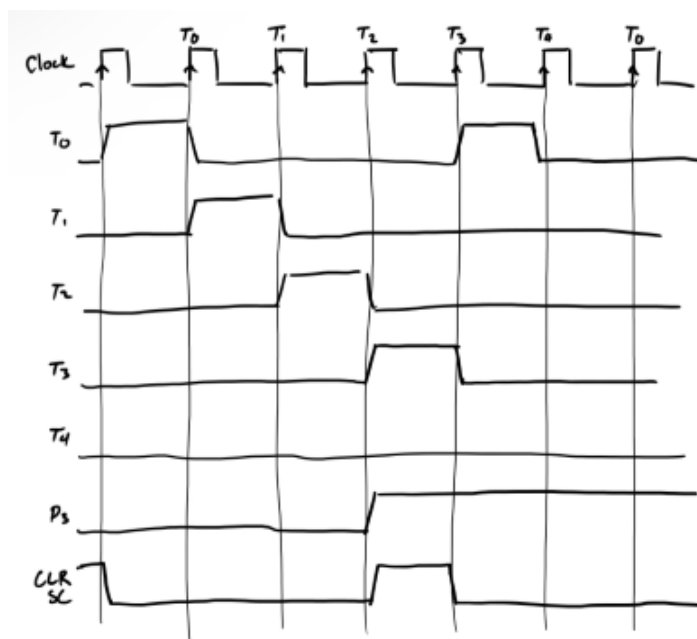
What are the two instructions needed in the basic computer to set the E flip-flop to 1?

Answer:

1. **CLA (Clear Accumulator)**: Clears the accumulator.
2. **CME (Complement E)**: Complements the E flip-flop, setting it to 1.

4. Problem 5-8 ~

Draw a timing diagram similar to Fig. 5-7 assuming that SC is cleared to 0 at time (T_3) if control signal (C_7) is active.



An attempt to explain

The timing diagram would show the following signals (from top to bottom):

1. Clock: Continuous square wave pulses through $T_0 \rightarrow T_4$
2. T_0 : Active during first time slot, then repeats after T_3 (due to early clear)
3. T_1 : Active during second time slot
4. T_2 : Active during third time slot

5. T_3 : Active briefly, then terminates due to C_7
6. T_4 : Would not become active due to SC being cleared at T_3
7. D_3 : Follows the same pattern as shown in Fig. 5-7 until T_3
8. CLR SC: Would show a pulse at T_3 when C_7 is active

Key differences from Fig. 5-7:

- The sequence terminates at T_3 instead of continuing to T_4
- A new cycle begins immediately after T_3 (shown by T_0 becoming active)
- The timing sequence becomes: $T_0 \rightarrow T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_0$ (skipping T_4)
- Total cycle length is shortened by one time unit